

Group

Q) Show that the set $\{1,2,3,4,5\}$ is not a group under addition and multiplication modulo 6

Q) Prove that the fourth roots of unity $1, -1, i$ and $-i$ form an abelian multiplicative group

Q) The set of all positive rational numbers under the composition defined by $a*b = (ab)/2$ is _____?

Q) Let $G = \{(a,b) \mid a,b \in \mathbb{R}, a \neq 0\}$. Define a binary operation $*$ on G by $(a,b) * (c,d) = (ac, bc+d)$ for all $(a,b), (c,d) \in G$. Is $(G, *)$ is a group?

Q) Construct the composition table for the set $T = \{1, 2, 3\}$ under a binary operation $(*)$ defined as follows:

$$1 * 1 = 1, 1 * 2 = 3, 1 * 3 = 2,$$

$$2 * 1 = 3, 2 * 2 = 1, 2 * 3 = 2,$$

$$3 * 1 = 2, 3 * 2 = 2, 3 * 3 = 1.$$

- Construct the composition (Cayley) table for $(T, *)$.
- Determine whether $(T, *)$ forms a group. Justify your answer.

Q) Determine whether the set of all 2×2 invertible matrices forms a group under matrix multiplication

Q) Let G be a group such that $(ab)^2 b = a^2 b^2$
for all $a, b \in G$. Is it **abelian group**?

- Q) If G is a group. Assume that a has order 5 and $a^3b = ba^3$. Is this abelian?

Q) Let M_2 be the set of 2×2 matrices over R i.e., $M_2 = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mid a, b, c, d \in R, ad - bc \neq 0 \right\}$. Is $(M_2, +)$ and $(M_2, \cdot)$ associative and commutative?

Q Construct the truth table for:

1. $p \rightarrow (q \vee r)$

2. $(p \vee q) \wedge (\neg p)$

3. $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$

4. $(p \vee q) \leftrightarrow (\neg p \rightarrow q)$

- Verify using a truth table:

i) $\neg(p \wedge q) \equiv (\neg p \vee \neg q)$

ii) $p \rightarrow q \equiv \neg p \vee q$

iii) $(p \vee (q \wedge r)) \equiv ((p \vee q) \wedge (p \vee r))$

Q1) Determine whether the set together with the binary operation is semi group, a monoid or neither?

a) $\mathbb{N}$, where $*$ is defined as ordinary addition

b) $\mathbb{Z}^+$, where $a*b = \max(a,b)$ for all $a,b \in \mathbb{Z}^+$

c) $\mathbb{Z}^+$, where $a*b = a$ for all $a,b \in \mathbb{Z}^+$

d) $S = \{1,2,3,6\}$ where $a*b = \gcd(a,b)$

e) $S = \mathbb{N} \times \mathbb{N}$, where $(a,b) * (a',b') = (aa',bb')$

Q2) A binary operation $*$ is defined on Z by $x*y = x + y - xy$, $x, y \in Z$. Is $(Z, *)$ a semi group?

Q3) The set of integers with respect to + i.e, $(\mathbb{Z}, +)$ is a cyclic group. Find the generator.

Q4) The multiplicative group $\{1, w, w^2\}$ is a cyclic group. Find generators.

Q5) The group $(G, +_6)$ is a cyclic group where $G = \{0, 1, 2, 3, 4, 5\}$. Find generators.

Q6) Find the inverse permutation

1	2	3	4	5
2	3	1	5	4

Q7

Consider $(\mathbb{R}^*, \bullet)$. Decide if the following sets form subgroups of $\mathbb{R}^*$.

1. $H = \{x \in \mathbb{R}^* \mid x = 1 \text{ or } x \text{ is irrational}\}.$

2. $K = \{x \in \mathbb{R}^* \mid x \geq 1\}.$

Solution

1. $H = \{x \in \mathbb{R}^* | x = 1 \text{ or } x \text{ is irrational} \}$.

No.

Counter example:

Consider $\sqrt{2} \in H$.

Further $\sqrt{2} \bullet \sqrt{2} = 2$ but $2 \notin H$

Therefore since there is no closure, $H \not\subseteq \mathbb{R}^*$. $\square$

2. $K = \{x \in \mathbb{R}^* | x \geq 1\}$.

No.

Counter example:

Consider $2 \in K$.

Note that $2^{-1} = \frac{1}{2}$, but $\frac{1}{2} \notin K$.

Thus the inverse of 2 does not exist.

Since $a^{-1} \nexists \forall a \in K$, $K \not\subseteq \mathbb{R}^*$. $\square$

- Q8)

Let $G = (M_{22}(\mathbb{R}), +)$.

Suppose you take this matrix s.t. the trace of the matrix is zero.

$$\text{Thus } H = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mid a + d = 0 \right\}.$$

Show that $H \leq G$.

Solution:

Note that $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ is the identity of G and it is clearly an element of H . Thus condition (1) of Theorem 2 is satisfied.

We will show that $g * h^{-1} \in H, \forall g, h^{-1} \in H$.

Let $\begin{bmatrix} a & b \\ c & d \end{bmatrix}, \begin{bmatrix} e & f \\ g & h \end{bmatrix} \in H$.

Then $a + d = 0$ and $e + h = 0$ since the two matrices belong to H .

Consider $\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} -e & -f \\ -g & -h \end{bmatrix} = \begin{bmatrix} a - e & b - f \\ c - g & d - h \end{bmatrix}$.

Then $(a - e) + (d - h) = (a + d) - (e + h) = 0 - 0 = 0$.

Hence $\begin{bmatrix} a - e & b - f \\ c - g & d - h \end{bmatrix} \in H$.

Thus $H \leq G$.

Let $(G, *)$ be a group then $H \subseteq G$ is a subgroup if and only if

The following conditions are satisfied:

1. The identity $e \in G$ is in H .
2. If $h_1, h_2 \in H$, then $h_1 * h_2 \in H$.
3. If $h \in H$ then $h^{-1} \in H$.

Q9) Let G be a group of order 56.

Find the **largest proper subgroup order**.

Q 10, 11

- How many generators are there of the cyclic group G of order 8?
 - $\gcd(a,8) = 1$
- How many generators in the cyclic group $(\{1,2,3,4,5,6\}, x_7)$
 - 1 $\rightarrow$ only gives 1
 - 2, 4, 6 $\rightarrow$ repeat early (do not generate all elements)

Q12) Let G be a multiplication group of all positive real numbers and R the additive group of all real numbers. Is G a subgroup of R ? give reasons

Solⁿ 12: **No**, G is not a subgroup of R because although $R^+ \subset R$ the operations are different. A subgroup must have the same binary operation as the group, but G uses multiplication while R uses addition.

Q 13

- i) Can abelian group have non abelian subgroup?
- ii) Can a non-abelian group have an abelian subgroup?
- iii) Can a non abelian group have a non abelian subgroup?

Solⁿ 13 i) : If a group G is abelian, then for all
 $a, b \in G$, $ab = ba$

Any subgroup $H \subseteq G$ uses the same operation,
so for all $x, y \in H$ $xy = yx$

Hence, every subgroup of an abelian group is
also abelian

Solⁿ 13 ii) A non-abelian group can definitely contain abelian subgroups.

Example:

The group S_3 is non-abelian.

But it has subgroups like:

- $\{e, (12)\}$
- $\{e, (123), (132)\}$

These are **abelian subgroups**.

Elements of S_3

We write permutations in **cycle notation**:

Identity:

- $e = (1)(2)(3)$

Swaps (transpositions):

- $(12), (13), (23)$

Cyclic permutations:

- $(123), (132)$ So total = 6 elements

Solⁿ13 iii) A non-abelian group can also have non-abelian subgroups.

Example: S_4 is non-abelian.

It contains a subgroup isomorphic to S_3 , which is also non-abelian. So non-abelian structure can persist inside subgroups.

Q14) Prove that each of the following set of matrices is group with respect to matrix multiplication

The set matrices of the form $\begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$;

$ad \neq 0, a, b, d \in R$. Is the group abelian?

Q15 Prove that each of the following set of matrices is group with respect to matrix multiplication

The set matrices of the form $\begin{pmatrix} \cos a & \sin a \\ -\sin a & \cos a \end{pmatrix}$

$a \in \mathbb{R}$.

Q16 Prove that each of the following set of matrices is group with respect to matrix multiplication

The set matrices of the form $\begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix}$

$c \in \mathbb{R}$.

Q17) Let $(S, *)$ be a commutative semi group. Show that if $x*x=x$ and $y*y=y$ then $(x*y)(x*y)=x*y$

Q18) Let $(G, *)$ be a group and $a, b \in G$.
Suppose that $a * b = b * a^{-1}$ and $b * a = a * b^{-1}$
show that $a^4 = b^4 = e$

Q19) Let $X=\{0,1,2,3\}$. Given $a,b \in X$ define $a*b=a$. Is this group, or semi group?

Q20) Let $X=\{0,1,2\}$. Given $a,b \in X$
define $a*b = c$ where c is the
remainder after $a+b$ is divided by 3
Find identity element, and inverse

- Q21)
 - i) For the set $I_4 = \{0,1,2,3\}$, Is modulo 4 system a field.
 - ii) Find whether modulo 5 system is a field for the set $\{0,1,2,3,4\}$.
 - iii) Show that the set $\{0,1,2,3,4,5,6\}$ is a field w.r.t addition modulo 7 and multiplication modulo 7 as the two composition.

Q22) In $(G, \times)$ $\times$ denotes multiplication numbers. Find whether it is abelian?

i) $G = \{2n : n \in \mathbb{Z}\}$

ii) $G = \{a+ib : a, b \in \mathbb{R}\}$

iii) $G = \{-1, 1\}$

Q23) Examine which of the following are groups. For those which fail to be groups mention which group axiom do not hold.

- a) The set R with respect to $*$ where $a*b = a$, for all $a \in R$
- b) The set Z with respect to $*$ where $a*b = a+b+1$ for $a, b \in Z$
- c) The set R with respect to, where $a*b = a+b-ab$ for all $a, b \in Z$
- d) The set Z with respect to where $a*b = |ab|$ for all $a, b \in Z$

Solution 23 a,b,c,d

(a)	No	No identity
(b)	Yes	Identity = -1 , inverse exists
(c)	No	Inverse fails for $a=1$
(d)	No	No identity

Q24) If $(\mathbb{R}, +)$ is a group of all real numbers under the operation '+' & $(\mathbb{R} - \{0\}, *)$ is another group of non-zero real numbers under the operation '*' (Multiplication) & f is a mapping from $(\mathbb{R}, +)$ to $(\mathbb{R} - \{0\}, *)$, defined as : $f(a) = 2^a$; $\forall a \in \mathbb{R}$.
Is f homomorphism?

Q25) Prove that H is a subgroup of the group $(C_0, \times)$ where $H = \{a+b\sqrt{2} \mid a \in \mathbb{Q}, b \in \mathbb{Q}, a^2+b^2 \neq 0\}$

Q26) If G is the multiplicative group of 3 cube-root units , i.e., $(G, o) = (\{1, w, w^2 \} , *)$ where $w^3 = 1$ & G' is an additive group of integers modulo 3 - $(G', o') = (\{1, 2, 3\} , +_3)$.

Is G isomorphism to G'